METRIC

A-A-54739
1 September 1992

COMMERCIAL ITEM DESCRIPTION

CASSEROLE, LABORATORY (Porcelain)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Federal Specification NNN-C-25B which is cancelled.

This Commercial Item Description covers glazed porcelain casseroles for laboratory use.

Salient Characteristics:

Shall be porcelain casseroles of the following types, classes, and milliliter capacities (see "Ordering data"):

Type I. Porcelain cup with porcelain handle and without cover.

Class 1. General laboratory use application.

Capacity (ml)
6
20
60
140
210
500
750
1200
2000

Class 2. Silica determination application

300 ml
350 ml

Type II. Porcelain cup with wooden handle and porcelain cover.

30 ml
150 ml
250 ml

AMSC\NA FSC 6640

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Materials.

Porcelain. The porcelain for the casserole cups and covers shall be vitreous ceramic whiteware having an average thermal coefficient of linear expansion from 20 to 200°C of 3.56 x 10^{-6} centimeter (cm)/cm/°C, based on the ASTM C372 test method for determination of linear thermal expansion.

Glaze coating. The glaze coating shall have a thermal coefficient of linear expansion comparable to that of the porcelain material. The glaze coating shall provide a permanent bonding to the porcelain, and provide protection against attack from silica material.

Wood handles. The handles for the Type II casseroles shall be shaped from hard wood, and shall be supplied with a protective finish.

Design and construction.

Type I casserole cups. The cups for type I casseroles shall be fabricated from porcelain and shall include integral porcelain handles. The construction of the class 1 and class 2 cups shall include the following:

Sloping and convex sides extending up from the bottom to the rim.
A pour-out spout on each cup that is located 90 degrees to the left of the handle, when the handle is positioned directly at the viewer.
The bottom of each cup is flat for stability.
The handle for each cup includes a flat sector for grasping, and a suitable cutout towards its end point for vertical suspension from a hook.

As a minimum, the cup and handle for each class 1 casserole shall be glazed inside and outside except for the rim.

As a minimum, the cup and handle for each class 2 casserole shall be glazed all over except for the outside bottom and tip of the handle.

Each casserole cup shall be permanently and legibly marked with the name or trademark of the manufacturer.

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Type II casserole cups. The cups for the type II casseroles shall be fabricated from porcelain and shall be supplied with an attached wood handle. The construction of the type II porcelain cups shall include the following:

The bottom of each cup is flat for stability.
Relatively straight sides that are rounded at the junction with the bottom.
The diameter at the bottom of each cup is slightly less than the diameter at the top.
A pour-out spout on each cup that is located 90 degrees to the left of the handle, when the handle is positioned directly at the viewer.

A wooden handle shall be securely attached to each cup in accordance with one of the following:

A handle with a threaded end is screwed into a threaded insert within a suitable integral projection from the porcelain cup.

A pin insert interconnects the shank end of the handle with a suitable insert within an integral projection from the porcelain cup.

A suitable heat-resistant thermosetting adhesive or epoxy cement is used to attach the handle to an integral projection from the porcelain cup.

As a minimum, the cup for each type II casserole shall be glazed inside and outside with the exception of the outside bottom and integral projection for attaching a handle.

Each casserole cup shall be permanently and legibly marked with the name and trademark of the manufacturer.

Type II casserole covers. The covers for the type II casseroles shall be fabricated from porcelain, and shall be glazed on all surfaces. The construction of the type II porcelain covers shall include the following:

Each cover has a circular shape with the overall diameter slightly larger than that for the top diameter of the mating casserole cup.

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The rim of each cover includes a projecting edge which fits over the rim of the mating casserole cup.

The top center of each cover includes a porcelain handle in the form of a molded knob.

Each casserole cover shall be permanently and legibly marked with the name or trademark of the manufacturer.

Dimensions and capacities. Casseroles shall be supplied with the dimensions and capacities indicated in table I.

Table I. Dimensions and capacities of casserole cups.

			DIMENSIONS (mm)	
		CAPACITY (ml)	TOP DIAMETER	HEIGHT
TYPE	CLASS	(+/- 15%)	(+/- 5 mm)	(+/- 5 mm)
I	1	6	31	15
I	1	20	43	23
I	1	60	70	35
I	1	140	89	45
I	1	210	100	50
I	1	500	133	77
I	1	750	144	90
I	1	1200	174	100
I	1	2000	183	135
I	2	300	110	60
I	2	350	117	56
II	-	30	50	25
II	-	150	88	43
II	-	250	107	53

Pour-out spout characteristics. The design of the pour-out spout for each casserole cup shall ensure delivery of an unbroken stream of liquid without wetting the outside wall of the cup.

Thermal resistance. The cups and covers for the casseroles shall be capable of withstanding temperatures of 1150°C.

Thermal shock resistance. The cups and covers for the type I and type II casseroles shall be able to exhibit no evidence of cracking as a result of being heated in an oven of 160 ± 2°C for 30 minutes, removed, and then plunged into a water bath with a temperature of 9 ± 1°C.

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Workmanship. The casserole cups and covers shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each (EA). One casserole, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Inspections and tests. Inspections and tests shall be conducted on the casserole cups and covers in accordance with commercial practice to determine compliance with specification requirements. Sampling for the inspections and tests shall be in accordance with commercial practice.

Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

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If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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Other publications. The following documents form a part of this specification to the extent that is specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposals shall apply.

ASTM C372 Standard Test Method for Linear Thermal Expansion of Porcelain Enamel and Glaze Frits and Fired Ceramics Whiteware Products by the Dilatometer Method.

(Applications for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103).

Ordering data. (Capacity size, type and class required, intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

NSN	CAPACITY	TYPE	CLASS	ITEM IDENTIFICATION
6640-00-290-6784	500 ml	I	1	CASSEROLE, LABORATORY
6640-00-412-8300	140 ml	I	1	CASSEROLE, LABORATORY
6640-00-412-8500	750 ml	I	1	CASSEROLE, LABORATORY
6640-00-492-5647	350 ml	I	2	CASSEROLE, LABORATORY
6640-00-511-6050	150 ml	II	-	CASSEROLE, LABORATORY
6640-00-551-9809	210 ml	I	1	CASSEROLE, LABORATORY
6640-00-893-6931	1200 ml	I	1	CASSEROLE, LABORATORY
6640-00-912-9090	20 ml	I	1	CASSEROLE, LABORATORY

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A-A-54739

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:
DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITY:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6640-1776

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:

1. DOCUMENT NUMBER: A-A-54739

2. DOCUMENT DATE (YYMMDD): 1 September 1992

3. DOCUMENT TITLE: CASSEROLE, LABORATORY (Porcelain)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME

b. ORGANIZATION

c. ADDRESS (Include Zip Code)

d. TELEPHONE (Include Area Code)

(2) AUTOVON

7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME
Defense Personnel Support Center
ATTN: DPSC-MSE

b. TELEPHONE (Include Area Code)
(1) Commercial: (215)7372870
(2) AUTOVON: 444-2870

c. ADDRESS (Include Zip Code)
2800 South 20th Street
Philadelphia, PA 19145

IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:
Defense Quality and Standardization Office
5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466
Telephone (703) 756-2340 AUTOVON 289-2340

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